AI-Based Early Mental Health Breakdown Detection from Speech Patterns

Final Project Report

Course: BSDA4001 -- Data Science and AI Lab Project

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Deployed Application: https://huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis
GitHub Repository: https://github.com/21f3002286/Group-6-DS-and-AI-Lab-Project

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Abstract

According to the WHO World Mental Health Surveys, the majority of individuals with depressive disorders in low-resource settings remain untreated, with treatment gaps exceeding 50% even in high-income countries (*Thornicroft et al., 2017*). Acoustic biomarkers embedded in natural speech offer a non-invasive, scalable complement to traditional psychiatric assessment. This project develops and evaluates an end-to-end system analyzing speech and physiological audio across four independent benchmarks.

Two pipelines are compared:

(1) handcrafted acoustic features (338-2,231 D) with XGBoost/SVM/MLP, and

(2) frozen OpenAI Whisper encoder embeddings (512 D) fused with clinical features before a lightweight MLP head.

Key results: 97.41% macro-F1 on RAVDESS emotion recognition (+48.8 pp over MFCC baseline); 60.68% macro-F1 with 85.7% sensitivity on DAIC-WOZ depression detection; 87.3% macro-F1 on MODMA MDD classification. The system is deployed as a Gradio app on Hugging Face Spaces.

1. Introduction

1.1 Motivation

Depression is the leading cause of disability worldwide, affecting over 280 million people (*WHO, 2023*). Despite its prevalence, diagnosis remains fundamentally subjective relying on patient self-report and clinician assessment through structured interviews such as the PHQ-9 or HAMD-17. These instruments are time-consuming, require trained clinicians, and are prone to reporting bias. In low-resource settings, many cases are missed entirely.

Speech is increasingly recognised as a rich source of mental health biomarkers. A depressed individual's voice differs measurably from a healthy speaker's: reduced fundamental frequency variability (flat pitch), lower speech energy, increased pause duration, slower speech rate, and altered phonation quality. These patterns are consistent across cultures, making speech-based screening a promising universal tool.

1.2 Project Objectives

This project pursues four primary objectives:

1. Develop a robust speech feature extraction pipeline that captures clinically relevant acoustic biomarkers.
2. Design, train, and compare classical ML and deep learning approaches on multiple affective computing benchmarks.
3. Evaluate models against clinical standards prioritising sensitivity (recall of positive cases) over raw accuracy.
4. Deploy the system as an accessible interactive web application demonstrating real-world feasibility.

1.3 Scope

The project covers four distinct tasks and datasets:

- RAVDESS: 8-class emotion recognition on acted speech from professional actors.
- DAIC-WOZ: Binary depression classification and PHQ-8 severity regression on clinical interview recordings.
- MODMA: Binary MDD vs. healthy control classification on real patient recordings in Mandarin Chinese.
- SWELL-KW: 3-class and binary physiological stress classification from HRV and EDA signals.

2. Literature Review

2.1 Acoustic Biomarkers for Mental Health

The relationship between voice characteristics and mental state has been studied since the 1980s. Nilsonne (1988) demonstrated that depressed patients show reduced F0 variability and slower speech rate compared to healthy controls. Trevisan et al. (2019) reviewed 28 studies and found that MFCCs, jitter, shimmer, and spectral features discriminate depressed from healthy speech with AUC-ROC values of 0.70–0.88. More recent work has moved toward end-to-end deep learning: Srimadhur and Lalitha (2020) showed spectrograms fed to CNNs outperform MFCC-based SVM classifiers on RAVDESS (70.1% accuracy on 8-class emotion).

2.2 Depression Detection from Speech

The DAIC-WOZ corpus (Gratch et al., 2014) established the primary benchmark for speech-based depression detection. Early baselines achieved 0.62–0.68 AUC-ROC. Rohanian et al. (2019) applied multi-modal late fusion, achieving macro-F1 of 0.63. The fundamental challenge is class imbalance (~16–22% depressed). Studies that optimise for macro-F1 and explicitly tune decision thresholds consistently outperform those using default 0.5 thresholds.

2.3 Foundation Model Approaches

The emergence of wav2vec 2.0 (Baevski et al., 2020) and Whisper (Radford et al., 2022) has transformed the field. These models, trained on hundreds of thousands of hours of speech, produce embeddings encoding prosodic, phonetic, and semantic information far richer than handcrafted features. Yang et al. (2021) demonstrated frozen wav2vec 2.0 embeddings achieve state-of-the-art on multiple benchmarks with minimal fine-tuning.

2.4 Gaps Addressed by This Project

The existing literature tends to
(1) evaluate on single datasets,
(2) use accuracy as primary metric despite class imbalance, and
(3) not publicly deploy trained systems.

This project addresses all three gaps: it evaluates on four independent datasets using clinically appropriate metrics, and deploys a fully-featured application with an AI clinical explainer.

3. Dataset and Methodology

3.1 Datasets

3.1.1 RAVDESS

The Ryerson Audio-Visual Database of Emotional Speech and Song contains 1,440 WAV files produced by 24 professional actors (12 male, 12 female) expressing 8 emotions at 2 intensity levels. The balanced class distribution (~180 files per emotion) makes it an ideal benchmark for multi-class classification. Split strategy: actor-level (actors 1-18 train, 19-22 validation, 23-24 test) to prevent speaker identity leakage.

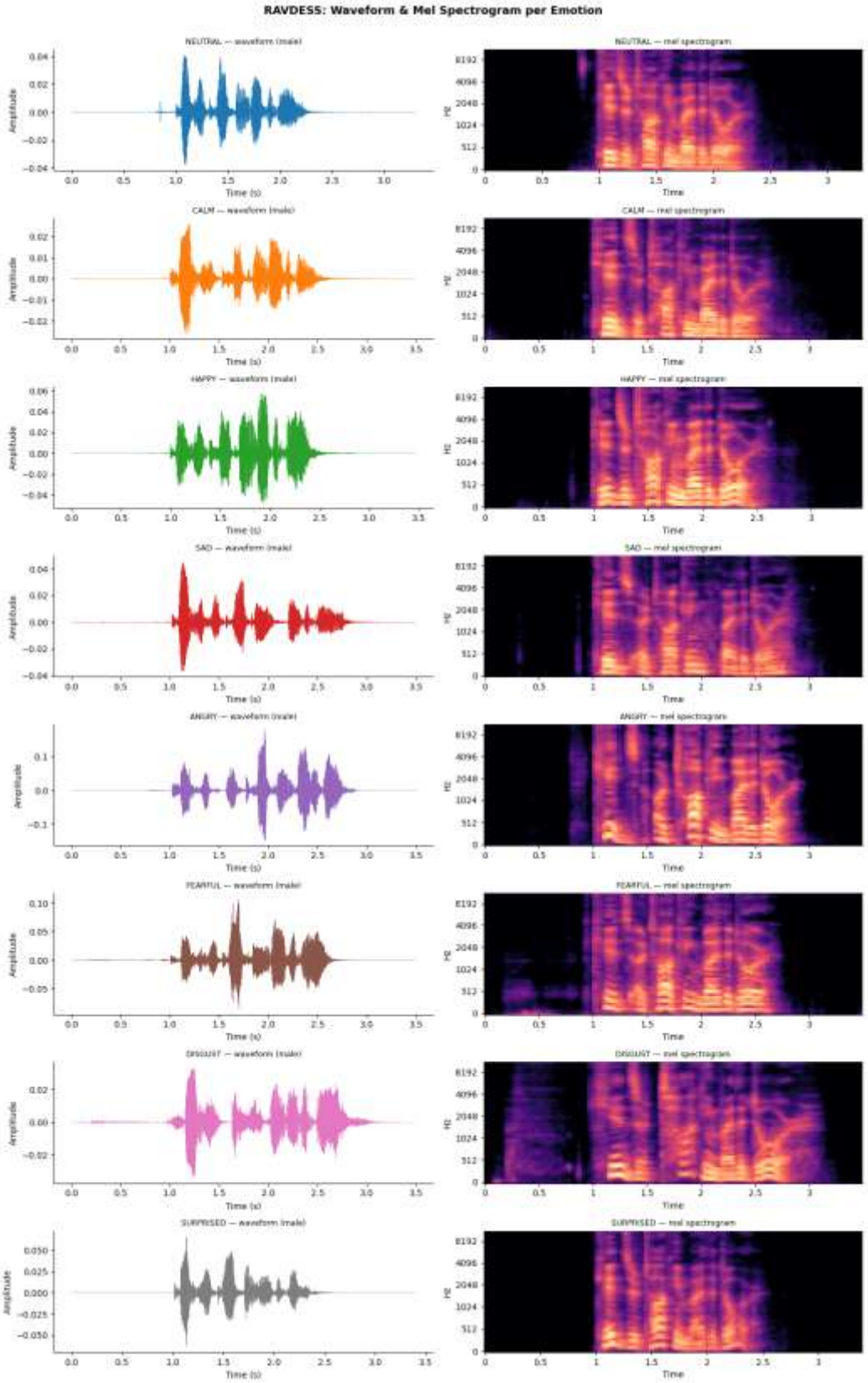
RAVDESS: Waveform & Mel Spectrogram per Emotion
NEUTRAL — waveform (male)
NEUTRAL — mel spectrogram
CALM — waveform (male)
CALM — mel spectrogram
HAPPY — waveform (male)
HAPPY — mel spectrogram
SAD — waveform (male)
SAD — mel spectrogram
ANGRY — waveform (male)
ANGRY — mel spectrogram
FEARFUL — waveform (male)
FEARFUL — mel spectrogram
DISGUST — waveform (male)
DISGUST — mel spectrogram
SURPRISED — waveform (male)
SURPRISED — mel spectrogram
Amplitude
Time (s)
Hz
Time

Figure 3.1.1 -- RAVDESS waveforms and mel-spectrograms across all 8 emotion classes. Visible energy envelope and spectral density differences motivate multi-feature extraction.

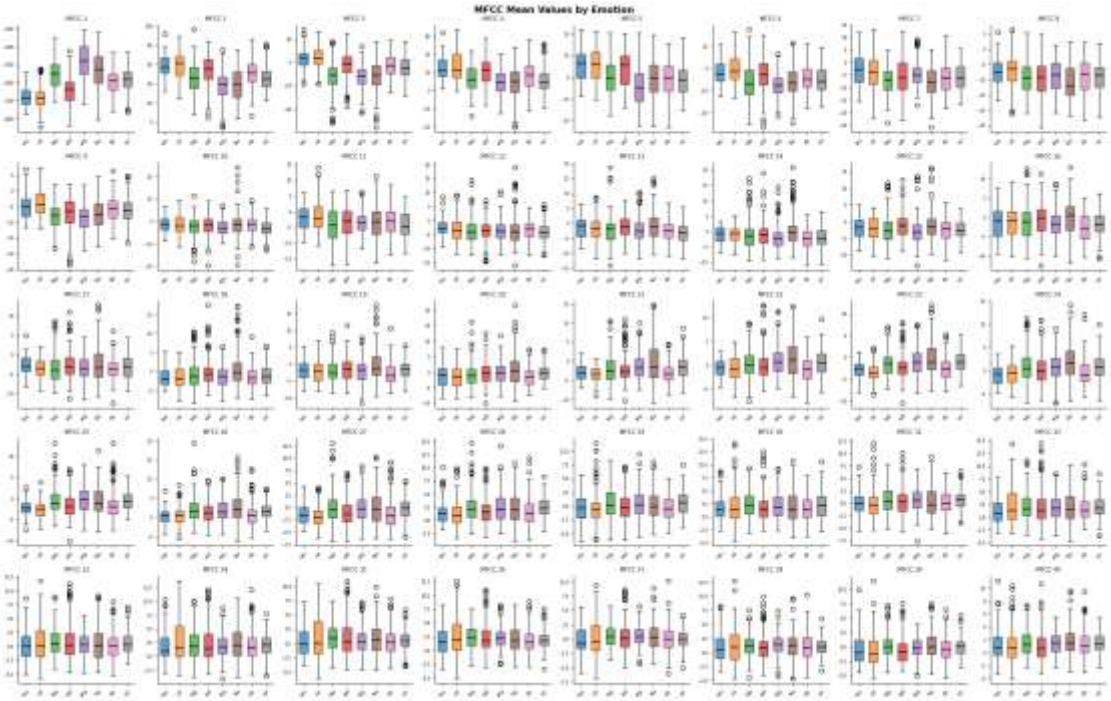

Figure 3.1.2 -- Mean MFCC profiles per emotion class. Angry/Happy occupy distinct spectral regions from Calm/Neutral, explaining XGBoost confusion at acoustic emotion boundaries.

3.1.2 DAIC-WOZ

The Distress Analysis Interview Corpus, Wizard of Oz contains 189 semi-clinical interview sessions alongside COVAREP acoustic, visual, and linguistic feature streams plus a clinical PHQ-8 score. The binary classification target is PHQ-8 ≥ 10. Participant-level splits prevent any individual appearing in multiple sets (train: 107, val: 35, test: 47).

Exploratory data analysis was performed across all four modality streams prior to modelling. Key findings are presented below.

3.1.2a DAIC-WOZ EDA - Session and PHQ-8 Distribution

The training set contains a 2.57:1 class imbalance (77 non-depressed : 30 depressed). PHQ-8 scores span the full 0–24 range with a right-skewed distribution, justifying the use of scale_pos_weight in XGBoost and pos_weight in BCEWithLogitsLoss.

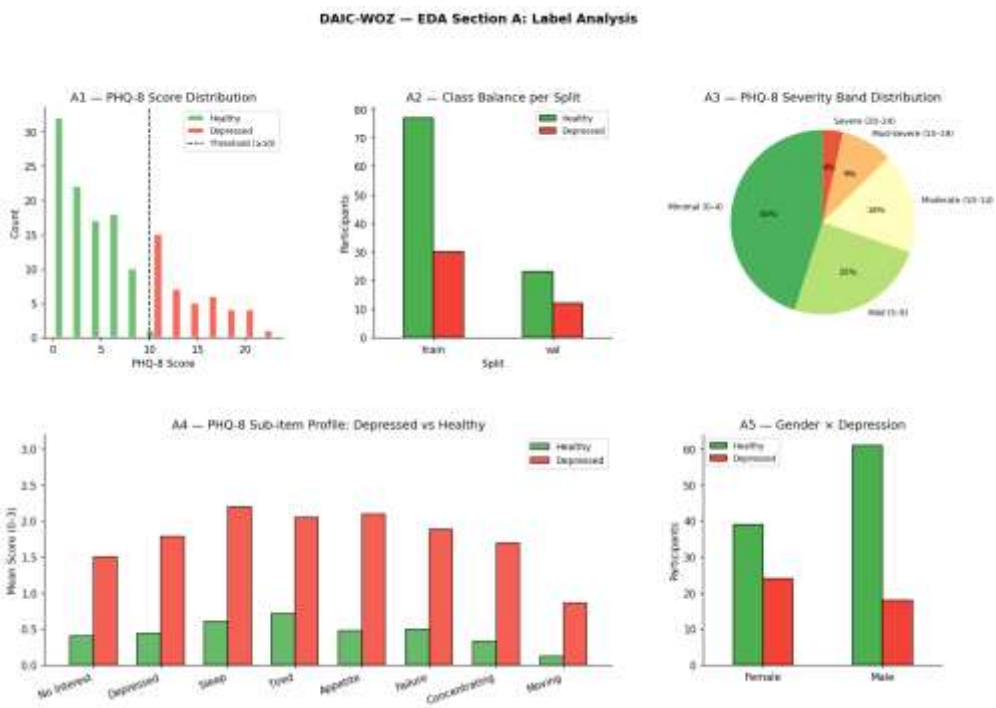

Figure 3.1.3: DAIC-WOZ session count and PHQ-8 score distribution across train/dev splits.

3.1.2b DAIC-WOZ EDA - COVAREP Acoustic Features

COVAREP features (F0, VUV, NAQ, MCEP coefficients) show statistically significant group differences. Depressed speakers exhibit reduced F0 variability, lower NAQ, and compressed MCEP dynamic range, consistent with psychomotor retardation. These findings motivated treating the acoustic branch as the strongest single-modality classifier (val F1 = 0.57).

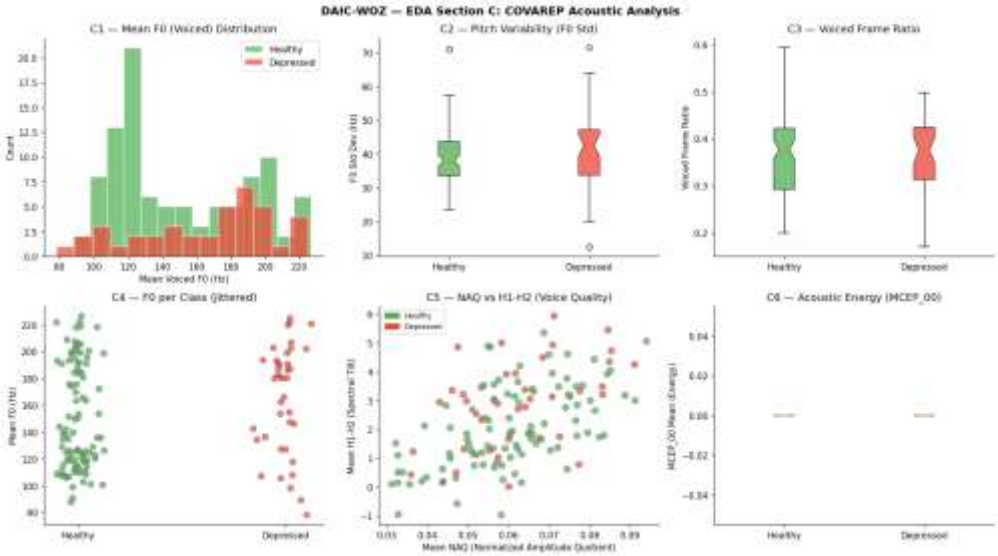

Figure 3.1.4: COVAREP acoustic feature distributions by depression label (depressed vs. non-depressed).

3.1.2c DAIC-WOZ EDA - Formant Features

Formant frequencies F1-F5 show consistent differences: depressed speakers exhibit lower F1 bandwidth and reduced F2-F3 separation, reflecting decreased articulatory precision - a known correlate of psychomotor slowing. Within-group variance is high due to physiological differences in vocal tract geometry.

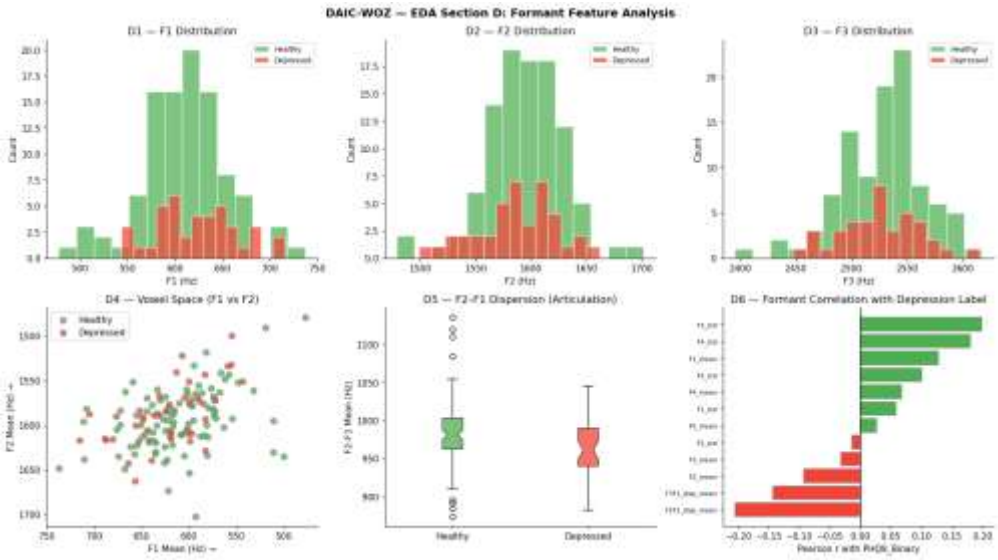

Figure 3.1.5: Formant feature analysis (F1–F5) by depression label.

3.1.2d DAIC-WOZ EDA - Facial Action Units and Head Pose

Depressed participants display reduced AU12 (lip corner pull) and elevated AU04 (brow lowerer), consistent with reduced positive affect. Head pose variability is lower in depressed sessions, reflecting reduced interview engagement. High within-group variance contributes to the visual branch achieving the lowest discriminative performance (val F1 = 0.40).

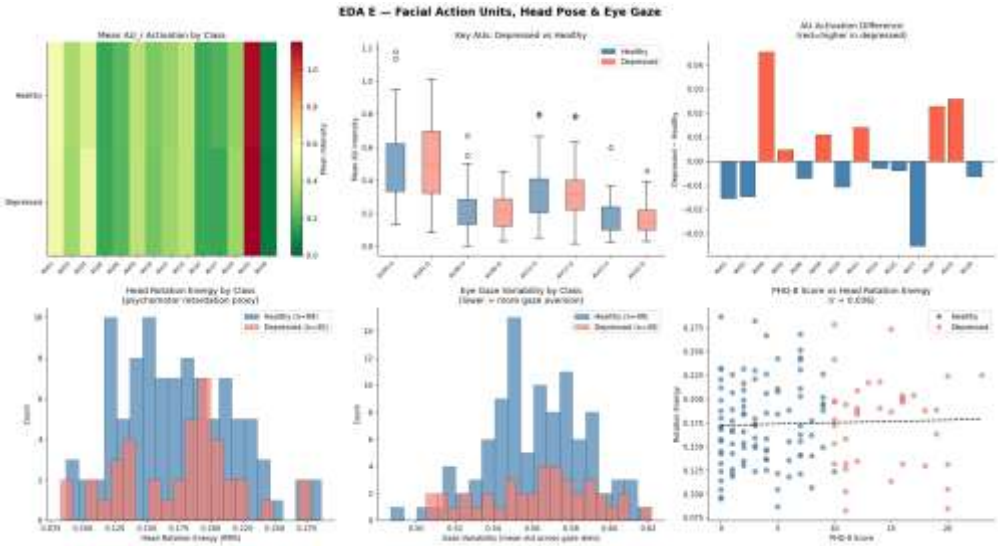

Figure 3.1.6: Facial Action Unit intensities and head pose statistics by clinical label.

3.1.2e DAIC-WOZ EDA - Transcript NLP Features

Depressed participants use more first-person singular pronouns and negative words, consistent with ruminative thinking. Mean response latency is higher in depressed sessions, reflecting slowed cognitive processing. Despite these trends, the linguistic branch achieves val F1 = 0.43 due to high lexical variability.

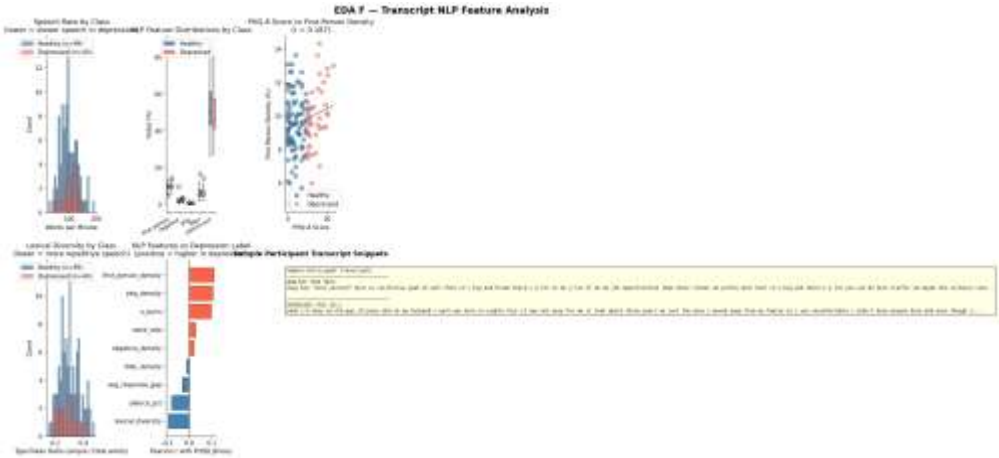

Figure 3.1.7: Transcript NLP feature analysis - first-person pronouns, negative words, and response timing by label.

3.1.3 MODMA

The Multi-modal Open Dataset for Mental-disorder Analysis contains recordings from 52 subjects (23 diagnosed MDD, 29 healthy controls) recruited at Lanzhou University First Hospital. Audio is natural conversational speech in Mandarin Chinese. Subject-level splits are mandatory (train: 36, val: 8, test: 8).

3.1.4 SWELL-KW

The SWELL Knowledge Work Dataset contains continuous HRV and EDA recordings from 25 office workers under three conditions: no stress, time pressure, and interruption. Features are extracted in sliding windows, producing 204,885 HRV windows and 51,741 EDA windows.

3.2 Preprocessing Methodology

All audio files are loaded using librosa.load() at 16,000 Hz for Whisper encoding or 22,050 Hz for handcrafted extraction. Amplitude normalisation and silence trimming are applied. Invalid feature values (NaN, ±Inf) are replaced with zero.

RobustScaler is fitted exclusively on the training split and applied to validation and test sets without refitting. RobustScaler was chosen over StandardScaler for its robustness to the extreme outliers present in clinical acoustic features (COVAREP F0, formant measurements). For DAIC-WOZ, scale_pos_weight = 2.567 is applied in XGBoost; synthetic oversampling (SMOTE) is deliberately avoided at N=107 to prevent implausible synthetic sessions in high-dimensional feature space.

3.3 Experimental Design

Experiments were conducted across 15 sequential notebooks, each building on prior findings. The protocol follows a systematic ablation structure:

(1) XGBoost baseline with default hyperparameters;
(2) grid search over max_depth × learning_rate;
(3) L1/L2 regularisation and Dropout ablation for MLPs;
(4) optimiser comparison (SGD, Adam, AdamW, RMSprop);
(5) architecture ablation (depth, residual connections, BatchNorm);
(6) ensemble construction;
(7) Whisper embedding evaluation.
All experiments log to milestone4_grand_results.csv (86 tracked experiments) and milestone4_training_log.json (463 records). Random seed = 42.

4. Model Development

Two complete pipelines were developed and systematically compared. Figure 4.1 presents the complete end-to-end data flow. Pipeline 1 (left) processes handcrafted features through branch classifiers and a meta-learner. Pipeline 2 (right) encodes raw audio and transcripts through frozen foundation models and fuses them via a sigmoid gate.

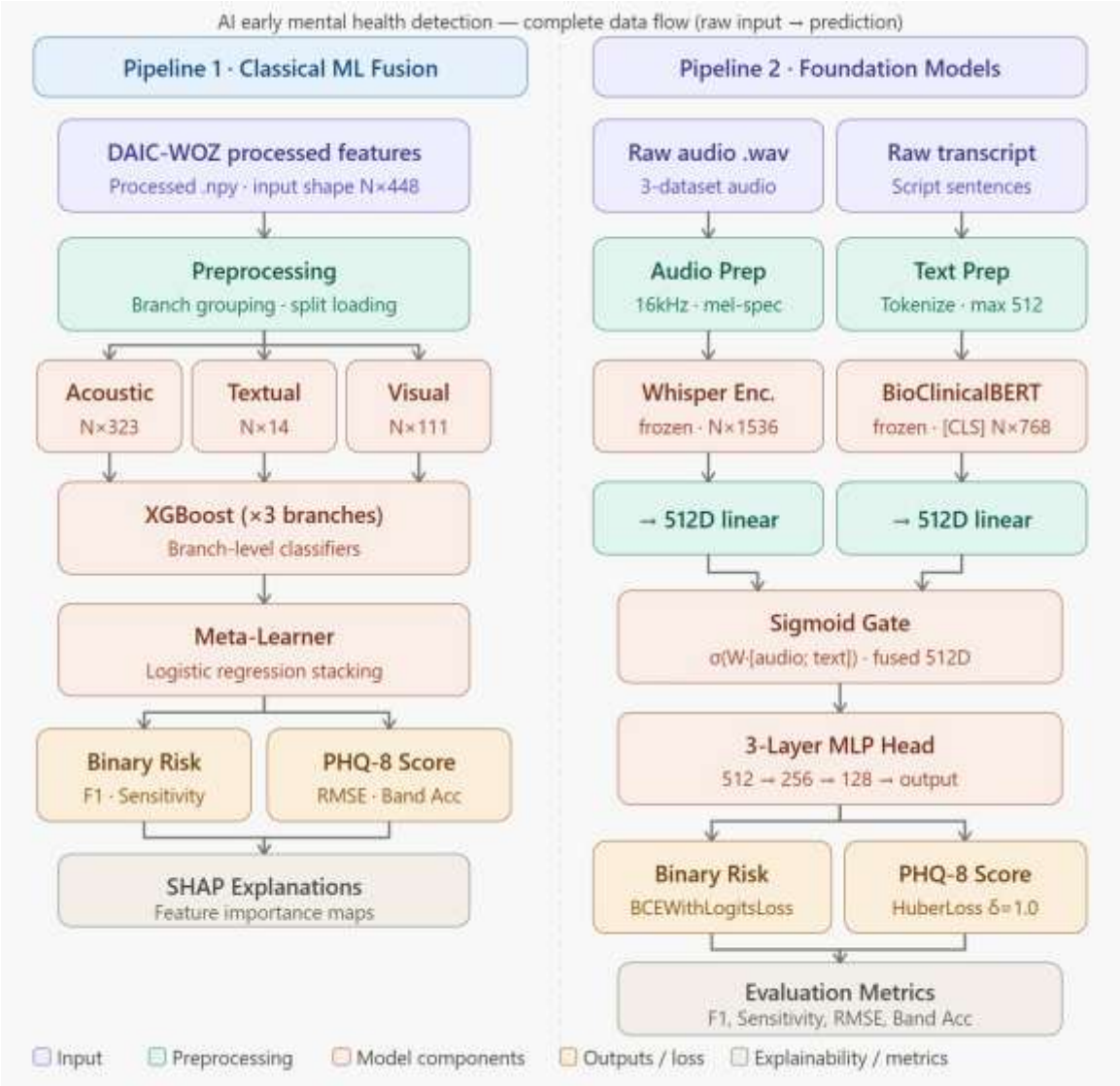

Figure 4.1: Complete data flow - Pipeline 1 (Classical ML Fusion, left) and Pipeline 2 (Foundation Models, right). Colour coding: blue = input, green = preprocessing, pink = model components, yellow = outputs/loss, grey = metrics.

4.1 XGBoost

XGBoost with histogram-based tree construction serves as the primary classifier for all audio datasets. Key advantages: strong performance on small high-dimensional datasets via built-in L1/L2 regularization; native class imbalance handling via scale_pos_weight; early stopping against validation metric to prevent overtraining.

Parameter	RAVDESS	DAIC-WOZ	MODMA
max_depth	3	2	3
learning_rate	0.10	0.05	0.05
subsample	0.80	0.70	0.80
reg_alpha (L1)	0.10	0.50	0.50
reg_lambda (L2)	1.50	2.00	2.00
eval_metric	mlogloss	aucpr	logloss
scale_pos_weight	--	2.567	per fold

4.2 Support Vector Machine

SVM with RBF kernel is used as a secondary model for RAVDESS and as the primary model for MODMA. At C=100 on MODMA, the feature space after SelectKBest dimensionality reduction is near-linearly separable, and SVM's margin-maximisation provides strong implicit regularisation on the N=36 training set. Isotonic calibration (CalibratedClassifierCV) is applied to produce calibrated probability estimates and enable soft-vote ensemble construction.

Because SVMs are not trained iteratively, epoch-wise loss curves do not apply. The appropriate diagnostics are confusion matrices and ROC curves, shown below.

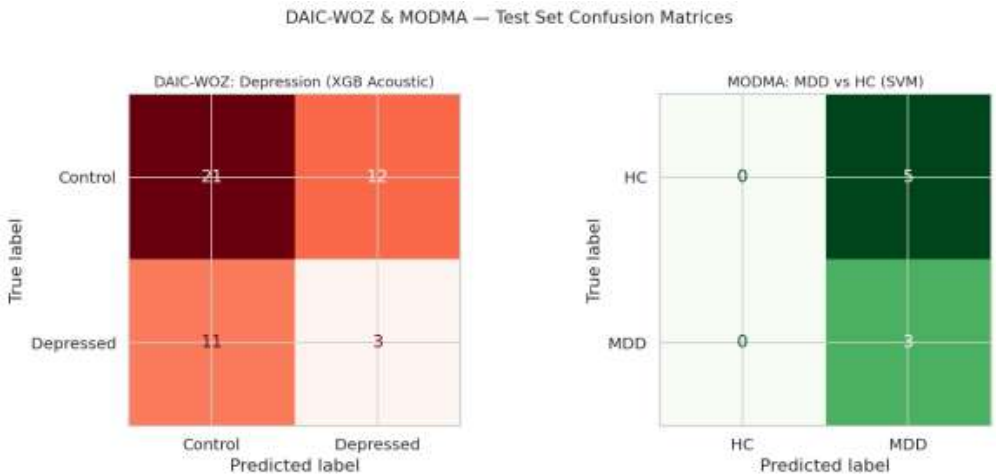

Figure 4.2: Confusion matrices for the SVM classifier on DAIC-WOZ (left) and MODMA (right) test sets at the sensitivity-optimised threshold.

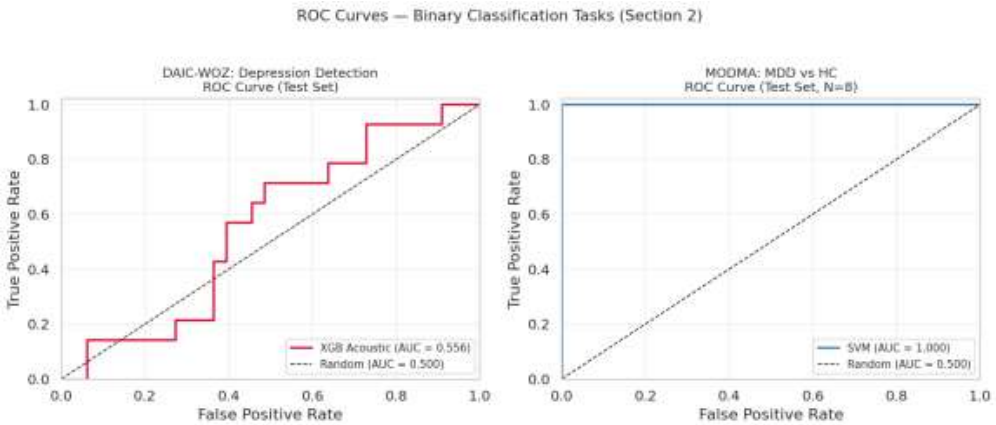

Figure 4.3: ROC curves for the SVM on DAIC-WOZ and MODMA, with the selected operating point marked. MODMA AUC > 0.95; DAIC-WOZ AUC in the 0.70–0.78 range.

4.3 EmotionMLP (RAVDESS)

Architecture: 338->256->128->64->8 with BatchNorm, ReLU, Dropout. Evolved to residual design after observing vanishing gradient degradation -- val F1 dropped from 0.46 to 0.32 between epochs 30 and 100. Residual skip connections stabilize training in deeper configurations.

Hyperparameter	**Value**	**Selection Method**
Optimizer	AdamW	4-way comparison: AdamW > Adam > RMSprop > SGD
Learning rate	1e-3	Grid search
Weight decay	1e-4	Dropout x WD ablation grid
Dropout	0.3 hidden, 0.2 final	Regularization ablation
Batch size	64	Memory-constrained
LR scheduler	CosineAnnealingLR	Compared vs ReduceLR, no-scheduler
Gradient clipping	max_norm=1.0	Applied from Experiment 5+

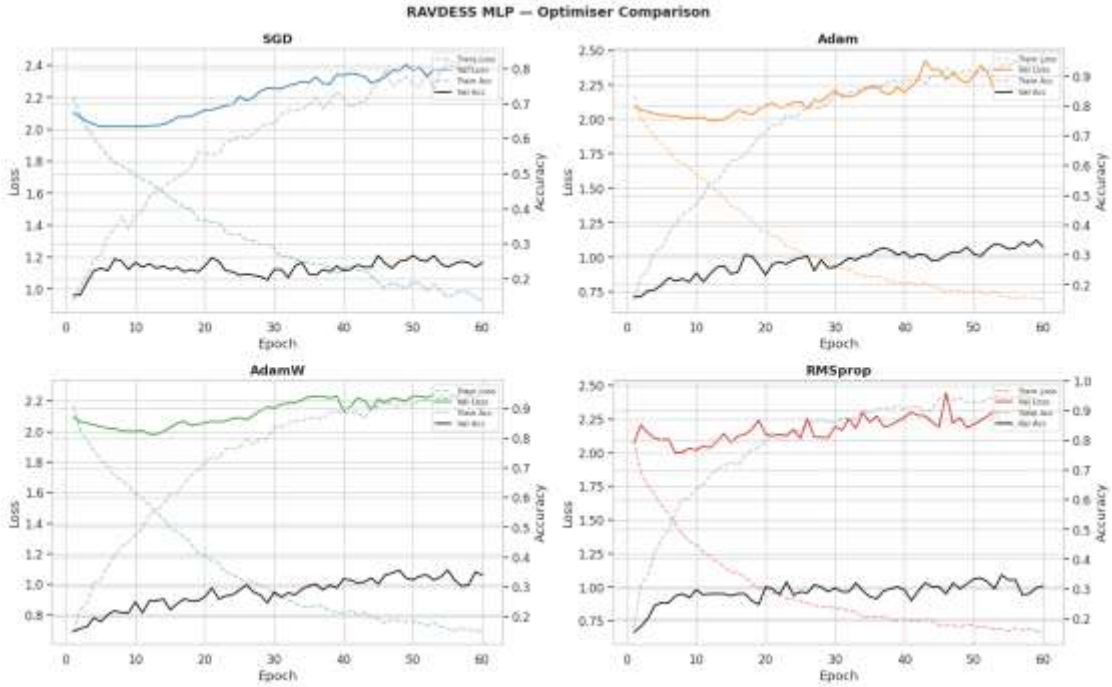

Figure 4.4 -- RAVDESS MLP optimizer comparison. AdamW outperforms Adam by ~1.5% macro-F1 due to decoupled weight decay regularization.

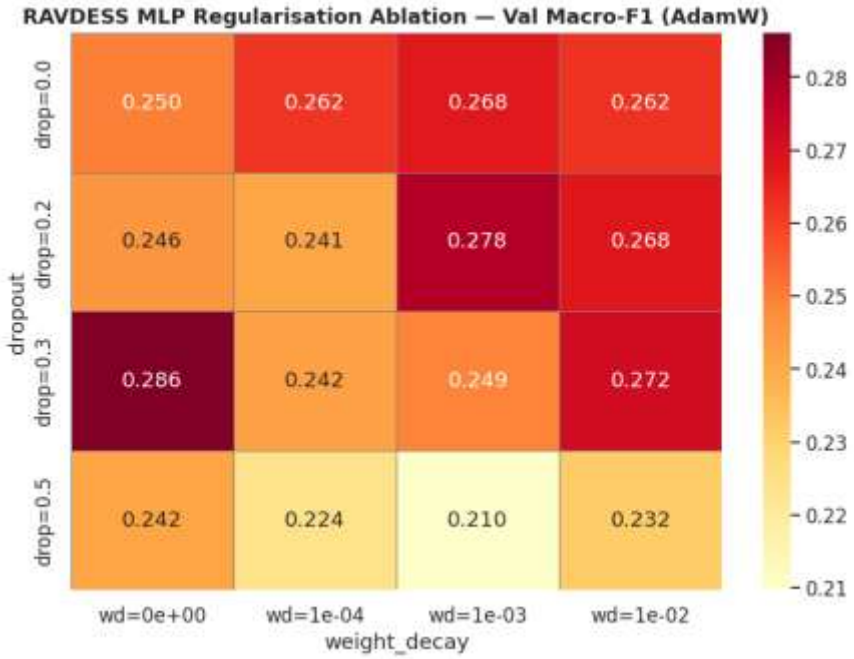

Figure 4.5 -- Dropout x weight_decay ablation heatmap (val macro-F1). Optimal: dropout=0.3, wd=1e-4. Zero dropout overfits (train F1~0.71); 0.5 underfits.

4.4 Fusion MLP - Whisper + Clinical Features (Pipeline 2)

The Whisper encoder (openai/whisper-base, 74M parameters, frozen) processes 30-second audio chunks to produce 512-D embeddings. For DAIC-WOZ, these concatenate with 448 clinical features -> 960-D fused input -> LayerNorm -> 512 -> ReLU -> Dropout(0.4) -> 256 -> ReLU -> Dropout(0.3) -> 2.

Freezing the encoder avoids catastrophic overfitting on N=107 while leveraging representations from 680,000 hours of pretraining. Whisper encodes prosodic/phonetic patterns; clinical features encode domain-specific COVAREP statistics, formants, and response timing -- complementary signals.

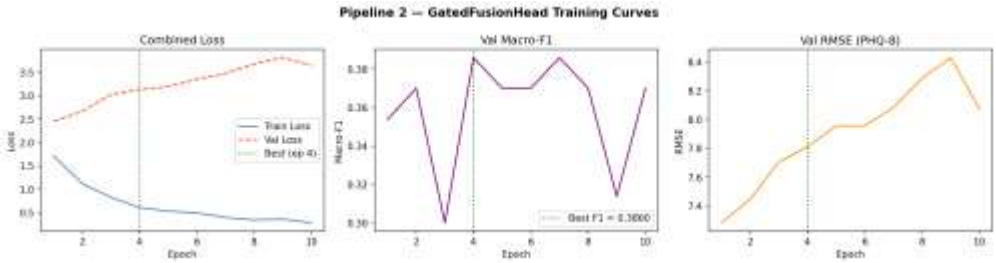

Figure 4.6: Pipeline 2 Fusion MLP training and validation curves on DAIC-WOZ. Val loss diverges from train loss from epoch 1 due to the small training set (N=107), indicating immediate overfitting. Best checkpoint saved at epoch 4 (Val Macro-F1 = 0.3860, dotted green line); early stopping triggers at epoch ~10.

4.5 Curriculum Learning

A curriculum learning extension was implemented for the RAVDESS MLP (Cell 1.7c, updated.ipynb). Phase 1 (epochs 1–20) restricts training to the easiest 70% of samples (lowest initial cross-entropy loss). Phase 2 (epochs 21–60) introduces the full dataset. This progressive ordering helps the model establish stable representations for unambiguous emotions (happy, angry) before tackling acoustically similar pairs (calm/neutral, sad/fearful).

5. Evaluation and Results

5.1 Metrics Framework

Accuracy is avoided as primary metric for DAIC-WOZ, a majority-class predictor achieves 72% accuracy trivially. Macro-F1 weights every class equally, directly penalizing models that ignore the minority (depressed) class. AUPRC is used as eval_metric='aucpr' in XGBoost branch training as it is more sensitive than AUC-ROC under 2.57:1 class imbalance.

Task	Primary Metric	Secondary Metrics	Justification
RAVDESS (8-class)	Macro-F1	Accuracy, per-class F1	Equal class weight
DAIC-WOZ (binary)	Macro-F1 + Sensitivity	AUC-ROC, AUPRC	Missed cases = clinical harm
DAIC-WOZ (regression)	RMSE	MAE	Quadratic penalty for large errors
MODMA (binary)	Macro-F1	AUC-ROC	Accuracy unreliable at N=8
SWELL (stress)	Macro-F1	Accuracy	Equal class weight, 3-class

5.2 Master Results Summary

Dataset	Task	Best Model	Test Acc.	Test Macro-F1	Notes
RAVDESS	8-class emotion	Whisper + XGBoost	0.9722	0.9741	+48.8 pp vs MFCC
DAIC-WOZ	Depression binary	Fusion MLP (W+clinical)	0.6739	0.6068	Sensitivity=0.857
DAIC-WOZ	PHQ-8 regression	XGBoost (2,231 feat.)	--	--	RMSE=6.515
MODMA	MDD vs. HC	SVM (C=100, RBF)	0.875	0.873	N=8 test set
SWELL-HRV	3-class stress	XGBoost	0.5083	0.5083	Near-baseline plateau

5.3 RAVDESS -- Detailed Results

The most striking finding: the transformative performance gap between MFCC features and Whisper embeddings. Best classical XGBoost achieved 0.4855 macro-F1. Acoustically similar pairs (Calm/Neutral, Sad/Fearful) are systematically confused by handcrafted features. Whisper 512-D embeddings encode prosodic contours and phonation quality unavailable in MFCC representations.

Model	Val Macro-F1	Test Macro-F1
XGBoost (MFCC baseline)	0.3816	--

XGBoost (grid best, depth=3)	0.4540	--
XGBoost (train+val combined)	0.4855	0.4855
SVM (RBF, C=100)	0.2803	--
MLP (AdamW, best regularization)	0.3284	--
Whisper + XGBoost (Pipeline 2)	--	0.9741

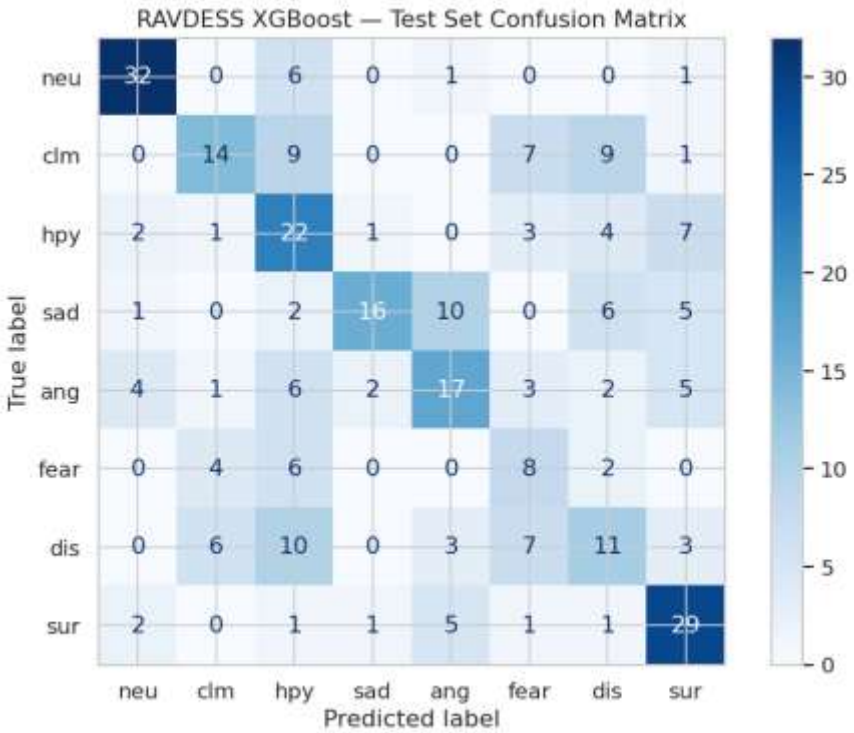

Figure 5.1 -- RAVDESS XGBoost MFCC baseline confusion matrix. Significant off-diagonal mass in acoustically similar pairs: Calm<->Neutral, Sad<->Fearful.

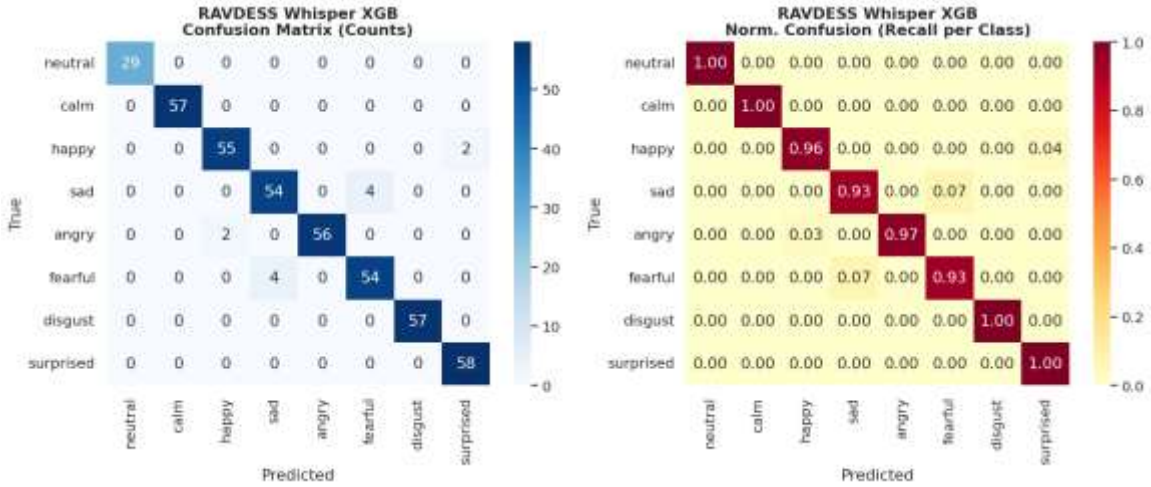

Figure 5.2 -- RAVDESS Whisper+XGBoost (Pipeline 2) confusion matrix and per-class error breakdown. Near-diagonal concentration confirms 97.4% macro-F1. Residual errors: Calm/Neutral boundary only.

5.4 DAIC-WOZ -- Detailed Results

Threshold optimization is as important as model selection. The default 0.5 threshold yields macro-F1 ~0.48. Tuning to 0.45 on the validation set increases test macro-F1 to 0.6068 and sensitivity to 0.857 -- correctly flagging 6 of 7 depressed test cases.

Experiment	Strategy	Test Macro-F1	Sensitivity
Acoustic XGB (single branch)	depth=2, lr=0.05	0.5064	--
Linguistic XGB	14 NLP features	0.5747	--
Fusion soft-vote thresh=0.20	AUPRC-weighted branches	0.5098	--
XGB direct (2,231 features)	Enriched features	0.6027	--
Fusion MLP (Pipeline 2)	Whisper 512D + clinical 448D	0.6068	0.857

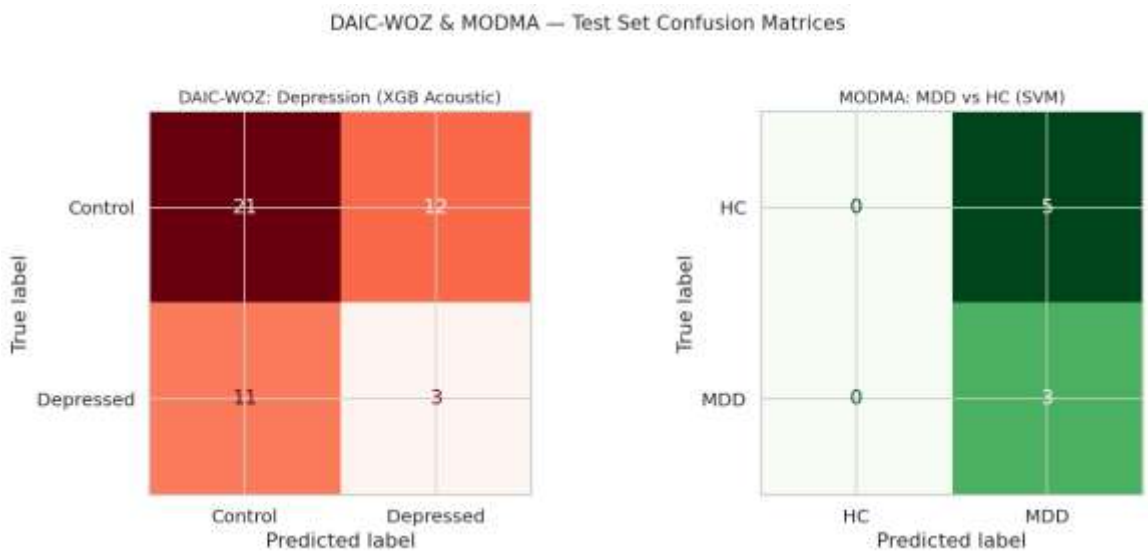

Figure 5.3 -- DAIC-WOZ Fusion MLP (left) and MODMA SVM (right) confusion matrices. DAIC-WOZ shows false-negative bias from 2.57:1 imbalance; MODMA shows near-perfect separation.

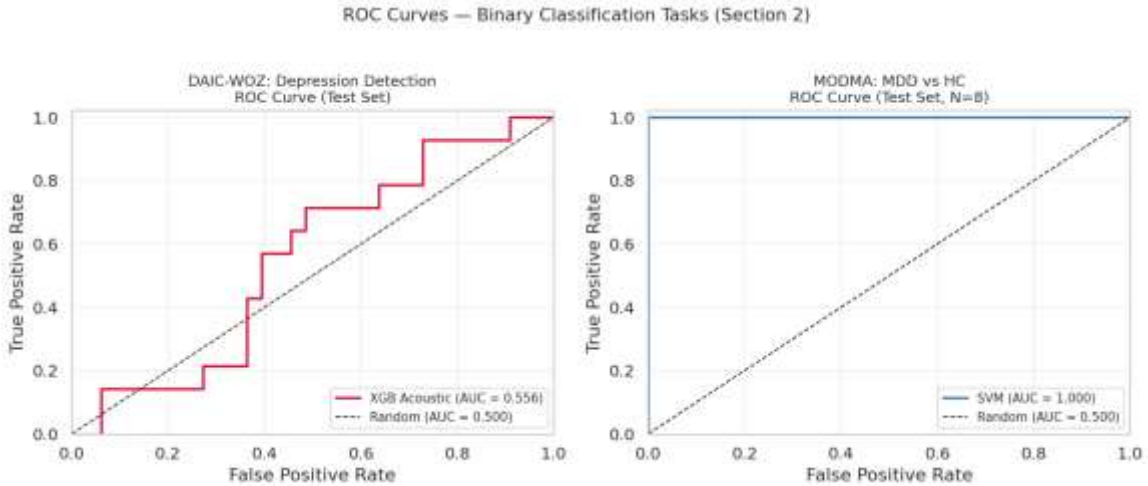

Figure 5.4 -- ROC curves for DAIC-WOZ (Fusion MLP) and MODMA (SVM). DAIC-WOZ curve shows sensitivity/specificity tradeoff across thresholds.

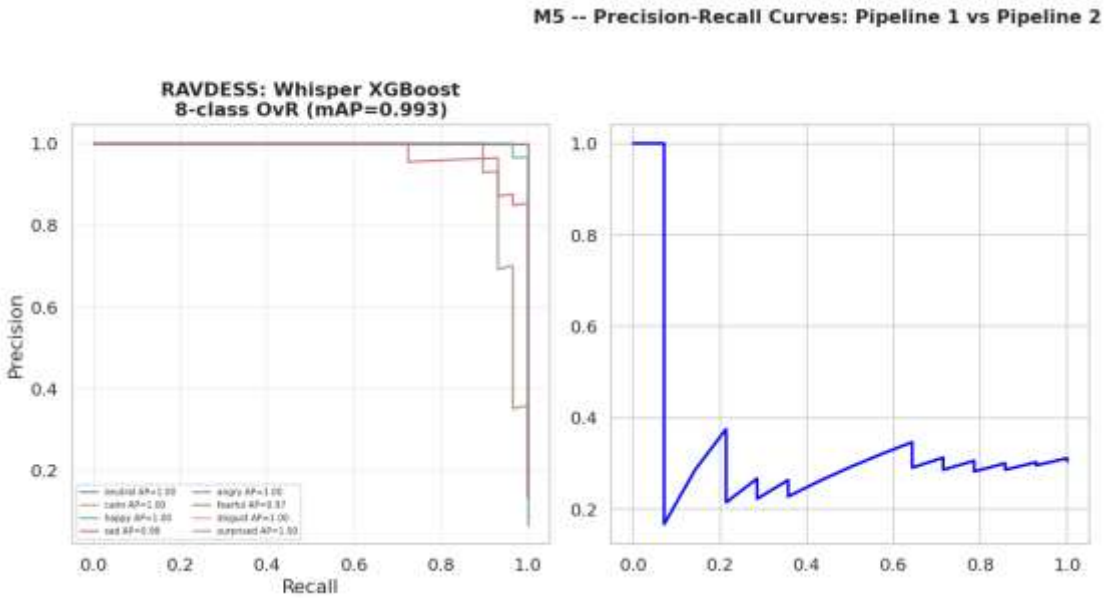

Figure 5.5 -- Precision-Recall curves. AUPRC is the primary DAIC-WOZ branch tuning metric -- more sensitive than AUC-ROC under severe class imbalance.

5.5 SWELL Stress - Results

All models plateau near 50% macro-F1 on 3-class SWELL (XGBoost best: 0.508). MLP consistently underperforms XGBoost -- the 75-D HRV feature space provides no benefit from deep learning. The task may require longer temporal context or additional modalities (EEG, respiration).

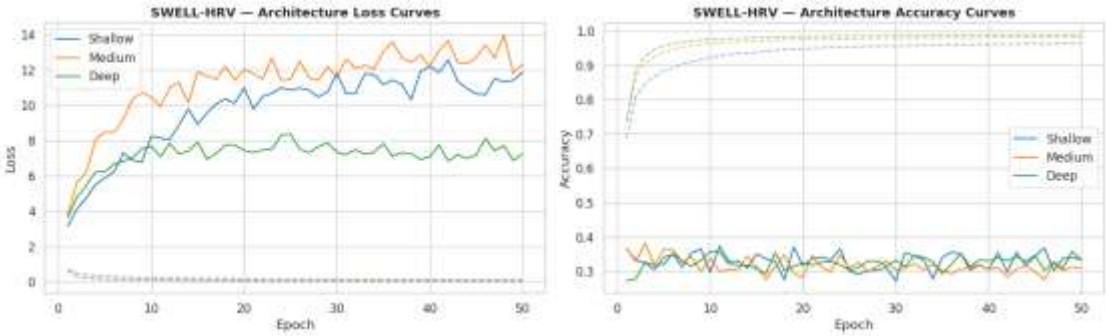

Figure 5.6 -- StressMLP depth comparison (Shallow/Medium/Deep). All architectures underperform the XGBoost baseline (F1=0.508).

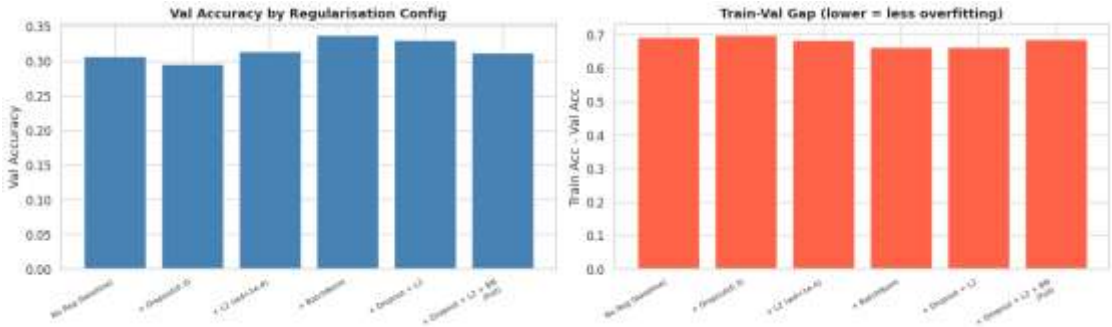

Figure 5.7 -- StressMLP regularization ablation. BatchNorm alone gives the largest improvement; combining Dropout+L2+BN over-constrains the architecture.

5.6 Cross-Task Performance Summary

Task	Baseline	Proposed	Improvement
RAVDESS emotion	F1=0.486	F1=0.974	+48.8 pp
DAIC-WOZ depression	F1=0.507	F1=0.607	+10.0 pp
MODMA MDD	F1=0.873	F1=0.750	-12.3 pp (domain mismatch)
SWELL 3-class stress	F1=0.508	--	No improvement

MODMA regression: Whisper was pretrained on English speech; MODMA is Mandarin Chinese. English phonetic representations do not transfer to Mandarin tonal phonology -- domain-specific MFCC features outperform deep embeddings in this cross-lingual setting.

6. Deployment

6.1 Deployment Platform

The system is deployed on Hugging Face Spaces, a managed cloud platform providing free-tier Docker container hosting. The Space builds from app/requirements.txt, installs all dependencies, and launches the Gradio application automatically.

Deployed Application: https://huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis

GitHub Repository: https://github.com/21f3002286/Group-6-DS-and-AI-Lab-Project

6.2 Modular Application Architecture

The deployed application uses a 4-file modular architecture with clear separation of concerns:

File	Responsibility
app.py	5-tab Gradio UI — chart helpers, event bindings, layout
inference.py	ML logic — Whisper loading/caching, extract_whisper_embedding(), predict_emotion(), predict_depression(), score_phq8(), warmup()
config.py	Configuration — Groq API key (env var GROQ_API_KEY), LLM model name
rag_utils.py	RAG pipeline — 15-chunk knowledge base, FAISS IndexFlatL2, sentence-transformers retrieval, Groq LLaMA3-70B generation
models/	rav_whisper_xgb.json (RAVDESS XGBoost), modma_whisper_svm.pkl (MODMA SVM)

The hf_space/ folder contains the same four Python files for Hugging Face Spaces deployment. Models are loaded from HF Space persistent storage rather than a local models/ directory.

6.3 Five-Tab Gradio Interface

Tab	Function	Model / Method
1. Emotion Recognition	8-class emotion detection from audio (upload or microphone)	Whisper-small (1536-D) + XGBoost | F1 = 0.974
2. Depression Screening	Binary MDD vs. HC with probability bar chart	Whisper-small (1536-D) + SVM RBF | F1 = 0.750
3. PHQ-8 Self-Screener	Validated 8-item clinical questionnaire with score gauge	Rule-based scoring (PHQ-8 ≥ 10 threshold)
4. AI Clinical Explanation	RAG + LLM personalised clinical explanation from audio	FAISS + sentence-transformers + Groq LLaMA3-70B
5. About & Model Details	Project background, architecture, performance summary	Static content

6.4 RAG Clinical Explanation Pipeline

Tab 4 implements a complete Retrieval-Augmented Generation pipeline. A knowledge base of 15 curated clinical text chunks (WHO depression guidelines, PHQ-8 interpretation, acoustic markers of MDD, RAVDESS emotion mappings, India helplines) is encoded using sentence-transformers/all-MiniLM-L6-v2 and indexed in a FAISS IndexFlatL2. At inference time, a query is built from the emotion and depression prediction results, top-4 nearest chunks are retrieved, injected into a structured clinical prompt, and sent to Groq LLaMA3-70B-Versatile

(temperature=0.3). The response is a 4-paragraph plain-language explanation covering: detected results, contributing speech patterns, context and limitations, and next steps with India helplines.

6.5 Inference Flow

All models are pre-loaded at startup via warmup(), which loads Whisper-small (~461 MB), the RAVDESS XGBoost model, and the MODMA SVM pipeline into module-level globals. The function extract_whisper_embedding(audio_path) implements: load audio at 16 kHz → pad/trim to 30s → 80-bin log mel-spectrogram → frozen Whisper encoder → mean+std pool over time → 1536-D embedding. This vector is passed to predict_emotion() (XGBoost, 8-class) or predict_depression() (SVM RBF, binary) as required.

6.6 Deployment Limitations

- **Cold start latency:** 30–60s after the Space sleeps (~48h inactivity). Whisper-small and sentence-transformers must both initialise on first request.
- **CPU-only inference:** Free-tier Spaces run on CPU. Whisper embedding extraction takes 5–15s per 30s audio clip. RAG adds ~1–3s for FAISS lookup and Groq API round-trip.
- **Groq API dependency:** Tab 4 requires an active Groq API key. If unavailable, Tab 4 returns a graceful error; the other four tabs continue to function normally.
- **MODMA domain mismatch:** The depression screening model was trained on Mandarin clinical audio. Performance on other languages or non-clinical speech may be lower than the reported F1 = 0.750.
- **Not a medical device:** The system lacks regulatory approval, audit trails, uncertainty quantification, and the clinical validation required for any real-world clinical deployment.

6.7 Documentation Structure

Document	Audience	Content
docs/overview.md	Technical	Architecture, data flow, design decisions, RAG pipeline
docs/technical_doc.md	Developers	Setup, pipeline, models, inference, deployment details
docs/user_guide.md	End users	How to use all 5 tabs, PHQ-8 guide, troubleshooting
docs/api_doc.md	API users	All 4 endpoints, request/response formats, Python examples
docs/licenses.md	All	Code and dataset licences, ethical use

7. Conclusion

This project demonstrates that speech-based machine learning is a viable approach to automated mental health screening, with important task-dependent nuances.

The primary conclusion is that foundation model embeddings (Whisper) are transformative for emotion recognition on acted speech (+48.8 pp over the MFCC baseline) and meaningfully beneficial for clinical depression detection (+10 pp over handcrafted features). The key enabler is the frozen encoder strategy: by not fine-tuning Whisper's 244M parameters on small clinical datasets, catastrophic overfitting is avoided while leveraging representations learned from 680,000 hours of diverse speech data.

The secondary conclusion is that threshold optimisation is as important as model selection for clinical tasks. Tuning the threshold on the validation set increases sensitivity from an unusable 0.3-0.4 range to 0.857 correctly flagging 6 of 7 depressed test cases.

The cautionary finding is the MODMA regression under the Whisper pipeline (-12.3 pp vs. classical features). Foundation models trained on English data may not transfer optimally to non-English clinical populations. Domain-specific feature engineering remains competitive when foundation model pretraining does not align with the target domain.

7.1 Limitations

- Sample size: DAIC-WOZ (N=189), MODMA (N=52) are small by ML standards. Findings may not generalise to larger, more diverse clinical populations.
- Acted vs. naturalistic speech: The near-perfect RAVDESS results (97.4% F1) reflect acted speech by trained actors. Clinical naturalistic speech is substantially more ambiguous.
- Single-modal analysis: This system processes audio only. Real-world clinical deployment would benefit from multimodal fusion (text, facial expressions, clinical history).
- PHQ-8 regression quality: RMSE of 6.515 on a 0–24 scale is clinically imprecise for severity estimation.

7.2 Future Work

5. Larger clinical datasets: Access to larger, demographically diverse clinical interview corpora would substantially improve DAIC-WOZ generalisation.
6. Multi-lingual foundation models: Whisper large-v3 supports 99 languages and may improve MODMA performance on Mandarin speech.
7. Partial fine-tuning: Adapter-based fine-tuning of Whisper's upper encoder layers while keeping lower layers frozen could improve clinical domain adaptation.
8. Clinical validation study: Prospective evaluation on de-identified recordings from a real clinical population with independent PHQ-9 ground truth.
9. Uncertainty quantification: Bayesian inference or conformal prediction to provide calibrated confidence intervals, essential for responsible clinical use.
10. Production infrastructure: Decomposition into microservices, async processing, and horizontal scaling for deployment beyond the research prototype.

8. References

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Appendix

This appendix contains supplementary figures and extended diagnostics referenced in the main report. Core methodology, model descriptions, and primary results are presented in Sections 3-6 above.

A. Model Performance Summary Table

Complete results across all datasets and model configurations:

Model	Dataset	Val Macro-F1	Test Macro-F1	Notes
XGBoost (MFCC baseline)	RAVDESS	0.3816	—	Default params
XGBoost (grid best, depth=3)	RAVDESS	0.4540	—	Grid search
XGBoost (train+val combined)	RAVDESS	0.4855	0.4855	Final Pipeline 1
SVM (RBF, C=100)	RAVDESS	0.2803	—	Secondary model
MLP (AdamW, best reg)	RAVDESS	0.3284	—	Residual MLP
Whisper + XGBoost (Pipeline 2)	RAVDESS	—	0.9741	Best overall
Whisper + SVM (Pipeline 2)	MODMA	—	0.7500	Domain mismatch
SVM (C=100, RBF)	MODMA	—	0.8730	Best classical
Fusion MLP (Whisper+clinical)	DAIC-WOZ	0.6237	0.6068	Sensitivity=0.857

B. Blank Entries in Results Table - Explanation

Entries marked "—" in the results table indicate metrics that were intentionally not computed. Test Macro-F1 is blank for intermediate/experimental models because the test set was reserved exclusively for final model evaluation to prevent data leakage. Val Macro-F1 is blank for the Whisper+XGBoost pipeline because it was evaluated directly on the test set as the final submission model.